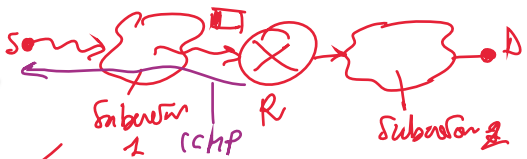
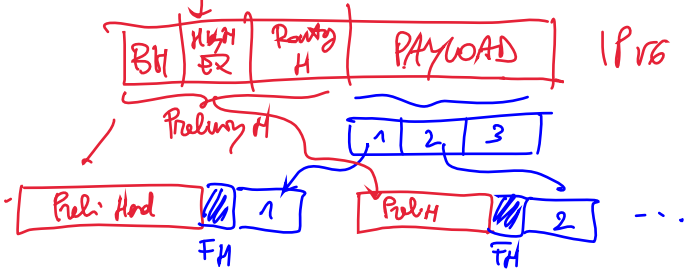
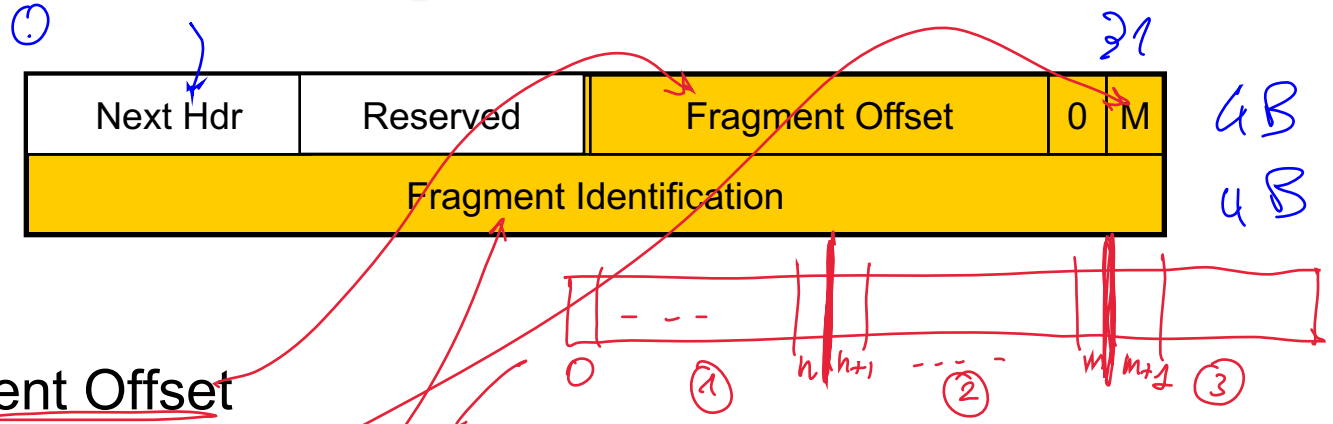
# Fragment Header




- Header of fixed length equal to 64bit
- Used for segmentation and reconstruction of datagrams
- In IPv6 only the source can fragment a datagram
  - **When an intermediate node receives a datagram with length longer than the MTU, it discards it and notifieds the error to the sender through ICMP**
- Every datagram is composed by a non-fragmentable and a fragmentable part
  - **The non-fragmentable part is formed by the header which need to be processed also in intermediate nodes (Basic Header, Hop by Hop Header e Routing Header)**
- All fragments but the last one have the same length multiple of 8 bytes

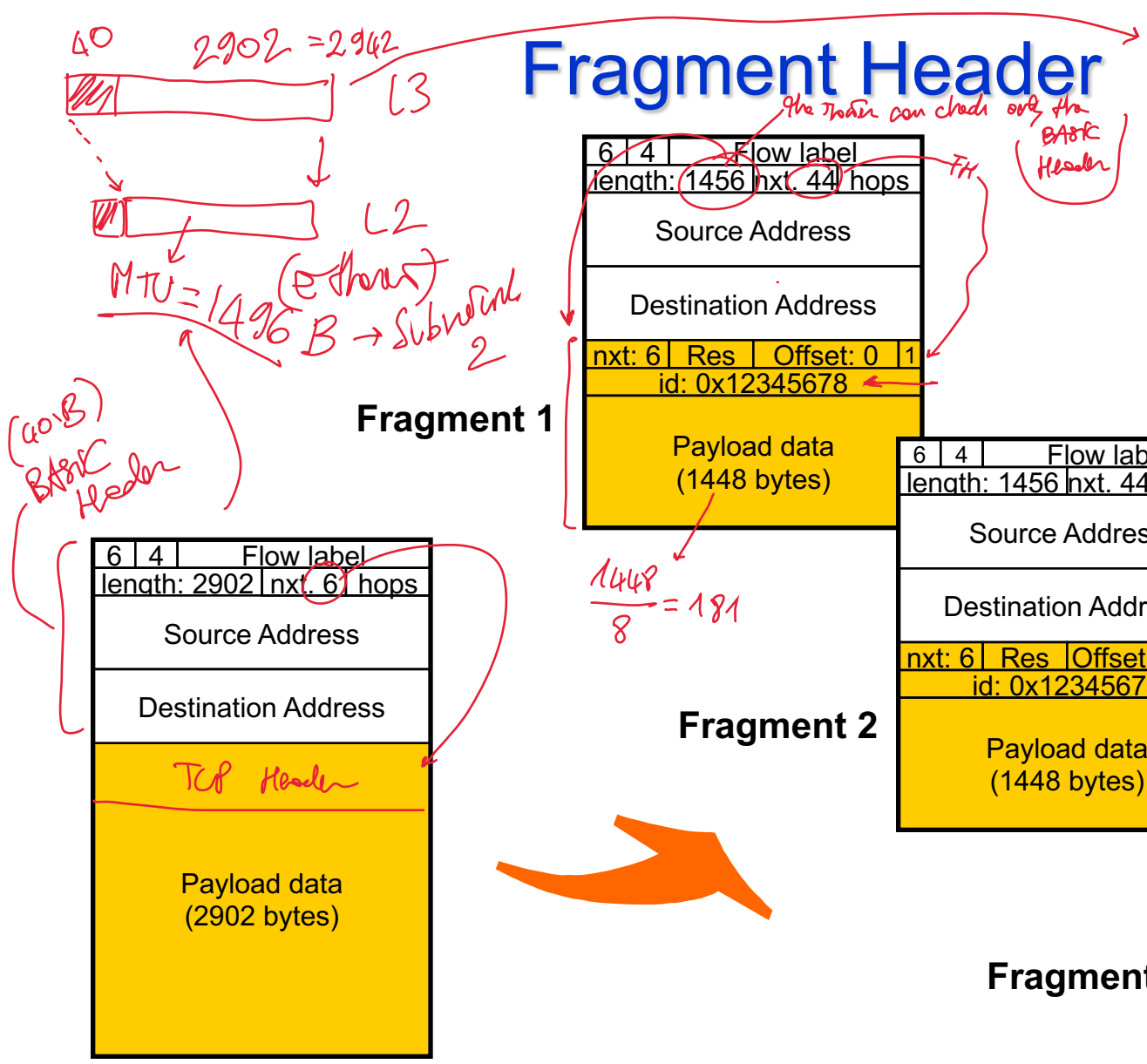


# Fragment Header




- Fragment Offset
  - **Specifies the position (in units multiple of 8 bytes) of the first byte of the fragment in the original datagram**
- Bit M (More Fragments)
  - **Its value is set to 1 in all fragments of the datagram but in the last one, where it is placed to 0**
- Fragment identification
  - **Identifies the datagram which the fragment belongs to**

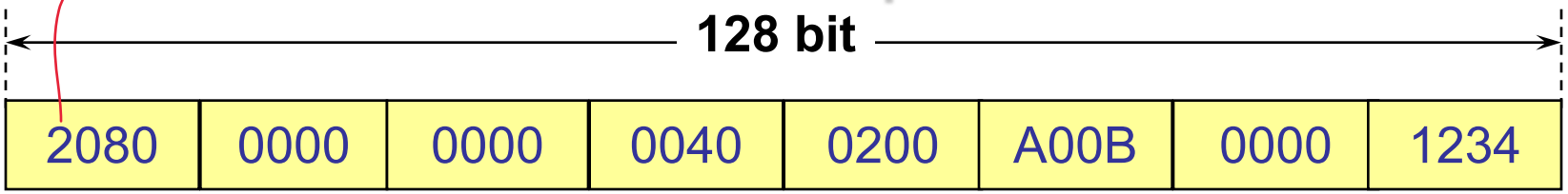
# Fragment Header



# IPv6 Addressing

- The designers of IPv6 had the benefit of IPv4 large experience and realized there would be tremendous advantages to designing the unicast address structure to reflect the overall topology of the Internet. These include:
  - **Easier allocation of address blocks at various levels of the Internet topological hierarchy**
  - **IP network addresses that automatically reflect the hierarchy by which routers move information across the Internet, allowing routes to be easily aggregated for more efficient routing**
  - **Flexibility for organizations like ISPs to subdivide their address blocks for customers**
  - **Flexibility for end-user organizations to subdivide their address blocks to match internal networks, much as subnetting did in IPv4**
  - **Greater "meaning" to IP addresses. Instead of just being a string of 128 bits with no structure, it would become possible to look at an address and know certain things about it**

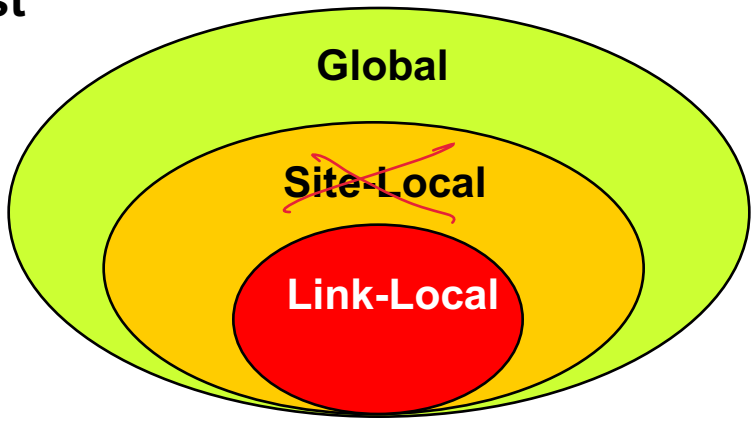
# IPv6 Address Representation



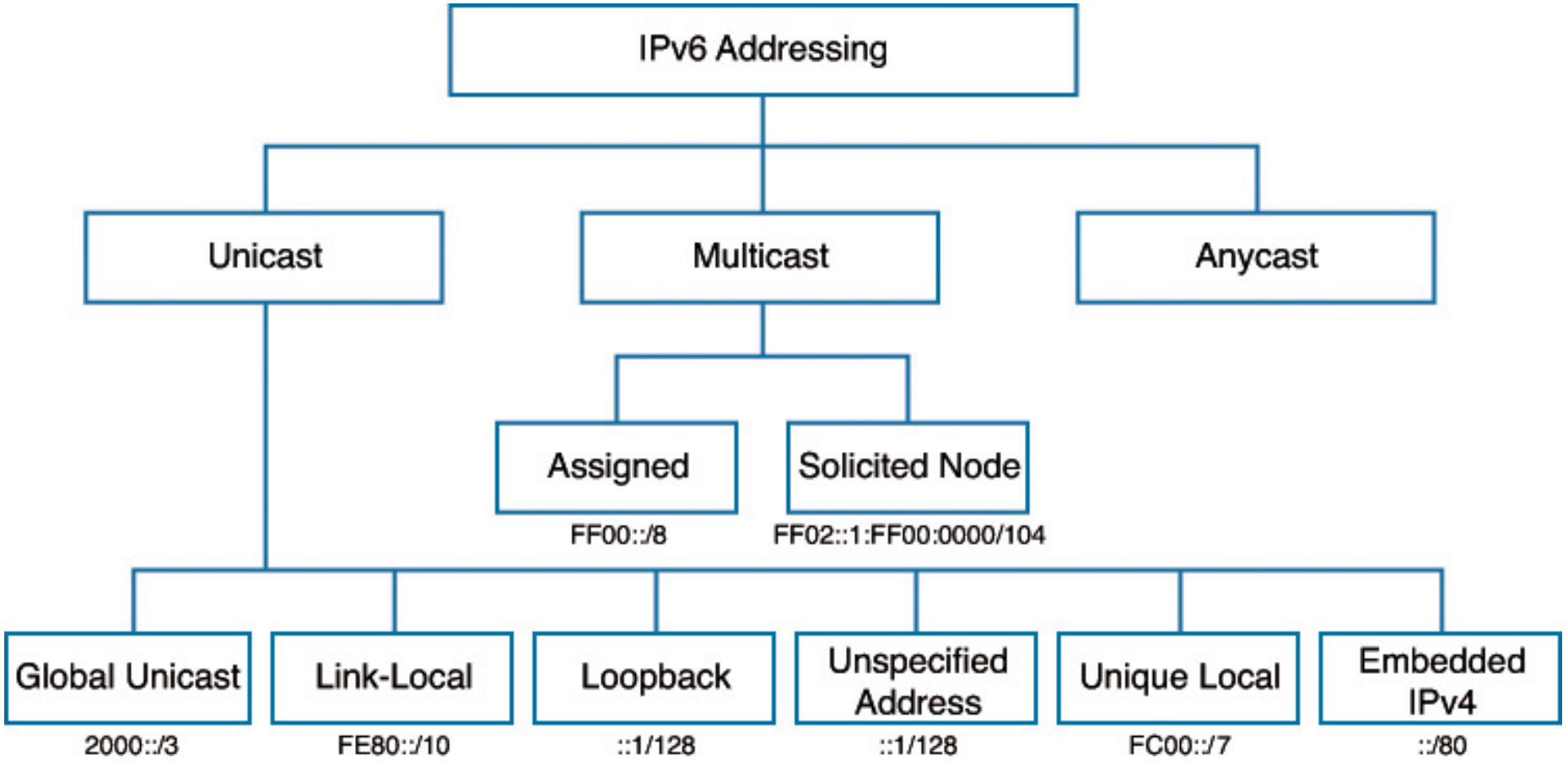

- 16-bit fields in case-insensitive colon hexadecimal (0,…,9,A,…,F) representation
  - 2080:0000:0000:0040:0200:A00B:0000:1234
- Some simplifications:
  - **leading zeroes in a field are optional and can be omitted**
    - 2080:0:0:40:200:A00B:0:1234
  - **successive fields of 0 represented as ::, but only once**
    - 2080::40:200:A00B:0:1234
    - 0:0:0:0:0:0:0:1 → ::1
    - 0:0:0:0:0:0:0:0 → ::
- IPv4-compatible address representation:
  - **0:0:0:0:0:0:10.0.0.1 = 0:0:0:0:0:0:A00:1 = ::A00:1 = ::10.0.0.1**

# IPv6 Addressing

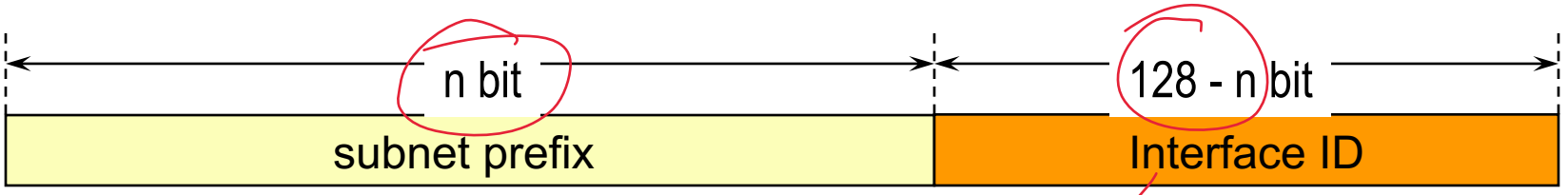
- IPv6 Addressing rules are covered by multiples RFC's
- Address Types are:
  - **Unicast : One-to-One (Global, Link local, Site local, Compatible)**
  - **Anycast : One to Nearest (Allocated from Unicast)**
  - **Multicast : One to Many**
  - **Reserved**
- A single interface may be assigned multiple IPv6 addresses of any type (unicast, anycast, multicast)
  - **No Broadcast Address → Use Multicast**
- Different address visibility
  - **Global**
  - **Site-Local (deprecated)**
  - **Link-Local**



# IPv6 Addressing



# IPv6 Addresses: Subnet Prefix




- Subnet prefix identifies:
  - **Type of address**
  - **Network/subnetwork that the address belongs to**
- The prefix length is specified by adding "/N" where N is the lenght (in bits)
  - **example:**
    - FEDC:0123:8700::1:2:3:4 /36
- Interface ID identifies:
  - **a node interface (within the given network)**

# IPv6 Addresses

- IPv6 addresses start with a variable length "Prefix Format" (PF) indicating the address type




| Address type | Binary prefix | IPv6 notation |
|---|---|---|
| Unspecified | 00...0 (128 bits) | ::/128 |
| Loopback | 00...1 (128 bits) | ::1/128 |
| Multicast | 1111 1111 | FF00::/8 |
| Link-local unicast | 1111 1110 10 | FE80::/10 |
| Site-local unicast | 11111110 11 | FEC0::/10 |
| Global unicast | 001 | 2000::/3 |

- Anycast addresses are taken by unicast addresses spaces

# IPv6 Addresses (cont.)

| *Binary prefix* | *Address type* |
|---|---|
| 0000 0000 | Reserved |
| 0000 0001 | Unassigned |
| 0000 001 | Reserved for NSAP (non-IP addresses used by ISO) |
| 0000 010 | Reserved for IPX (non-IP addresses used by IPX) |
| 0000 011 | Unassigned |
| 0000 1 | Unassigned |
| 0001 | Unassigned |
| 001 | Unicast Address Space |
| 010 | Unassigned |
| 011 | Unassigned |
| 100 | Unassigned |
| 101 | Unassigned |
| 110 | Unassigned |
| 1110 | Unassigned |
| 1111 0 | Unassigned |
| 1111 10 | Unassigned |
| 1111 110 | Unassigned |
| 1111 1110 0 | Unassigned |
| 1111 1110 10 | Link Local Use addresses |
| 1111 1110 11 | Site Local Use addresses |
| 1111 1111 | Multicast addresses |

# Reserved Addresses with prefix 0x00

## Unspecidfied address (only as source address)

| 8 bits | 120 bits |
|---|---|
| 0x00 | All 0s |

::0

## Loopback address

| 8 bits | 120 bits |
|---|---|
| 0x00 | 000000000000..... 000000000001 |

::1

## IPv4 compatible address (to send a pkt to a IPv6 host through IPv4 nets)

| 8 bits | 88 bits | 32 bits |
|---|---|---|
| 0x00 | All 0s | IPv4 address |

::x.y.z.w

## IPv4 mapped address (to send a pkt to a IPv4 host through IPv6 nets)

| 8 bits | 72 bits | 16 bits | 32 bits |
|---|---|---|---|
| 0x00 | All 0s | All 1s | IPv4 address |

::FFFF:x.y.z.w

# Unspecified and Loopback Addresses

- Unspecified address 0:0:0:0:0:0:0:0
  - **::**
  - **It must never be assigned to any node**
  - **It indicates the absence of an address**
  - **e.g. as src addr of any pkts sent by an initializing host**
  - **must not be used as the dest addr of pkts**
  - **a packet with a src addr :: must never be forwarded by an IPv6 router**
- Loopbackaddress 0:0:0:0:0:0:0:1
  - **::1**
  - **It may be used by a node to send an IPv6 packet to itself**
  - **It must not be assigned to any physical interface**
  - **It is treated as having link-local scope (of a virtual interface called "loopback interface")**
  - **must not be used as src addr in pkts sent outside of a single node**
  - **a packet with dest addr of loopback must never be sent outside**

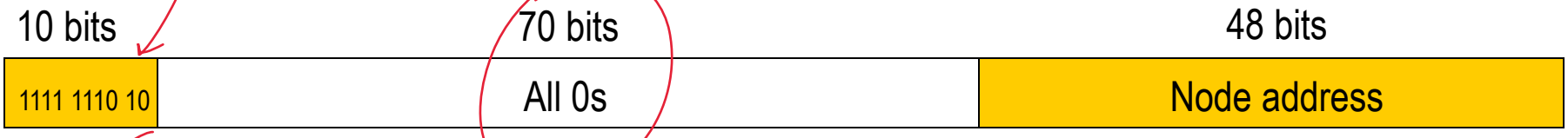
# IPv6 Addresses with Embedded IPv4 Addresses

- IPv4-Compatible IPv6 Address
  - **was defined to assist in the IPv6 transition (to send a pkt to a IPv6 host through IPv4 nets)**
  - **is now deprecated because the current IPv6 transition mechanisms no longer use these addresses**

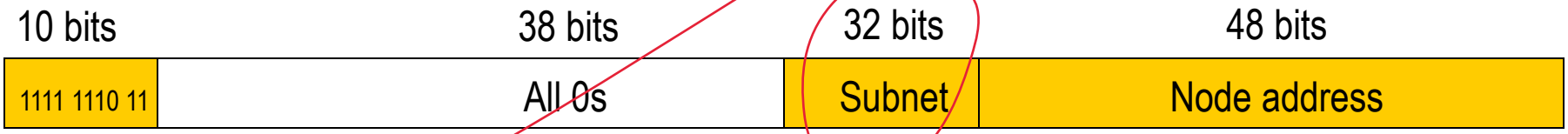
- IPv4-Mapped IPv6 Address
  - **IPv6 address that holds an embedded IPv4 address is defined**
  - **it is used to send a pkt to a IPv4 host through IPv6 nets**

# Local Addresses

- Link local address (only within a subnet, e.g. LAN)



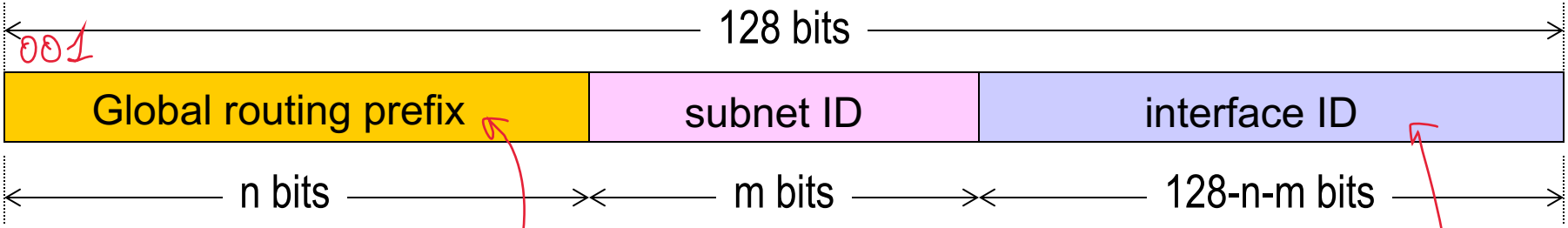

- Site local address (only though private subnets)




- Note:
  - **Site-local addresses were originally designed to be used for addressing inside of a site without the need for a global prefix**
  - **Site-Local addresses are now deprecated**

# Global Unicast Addresses

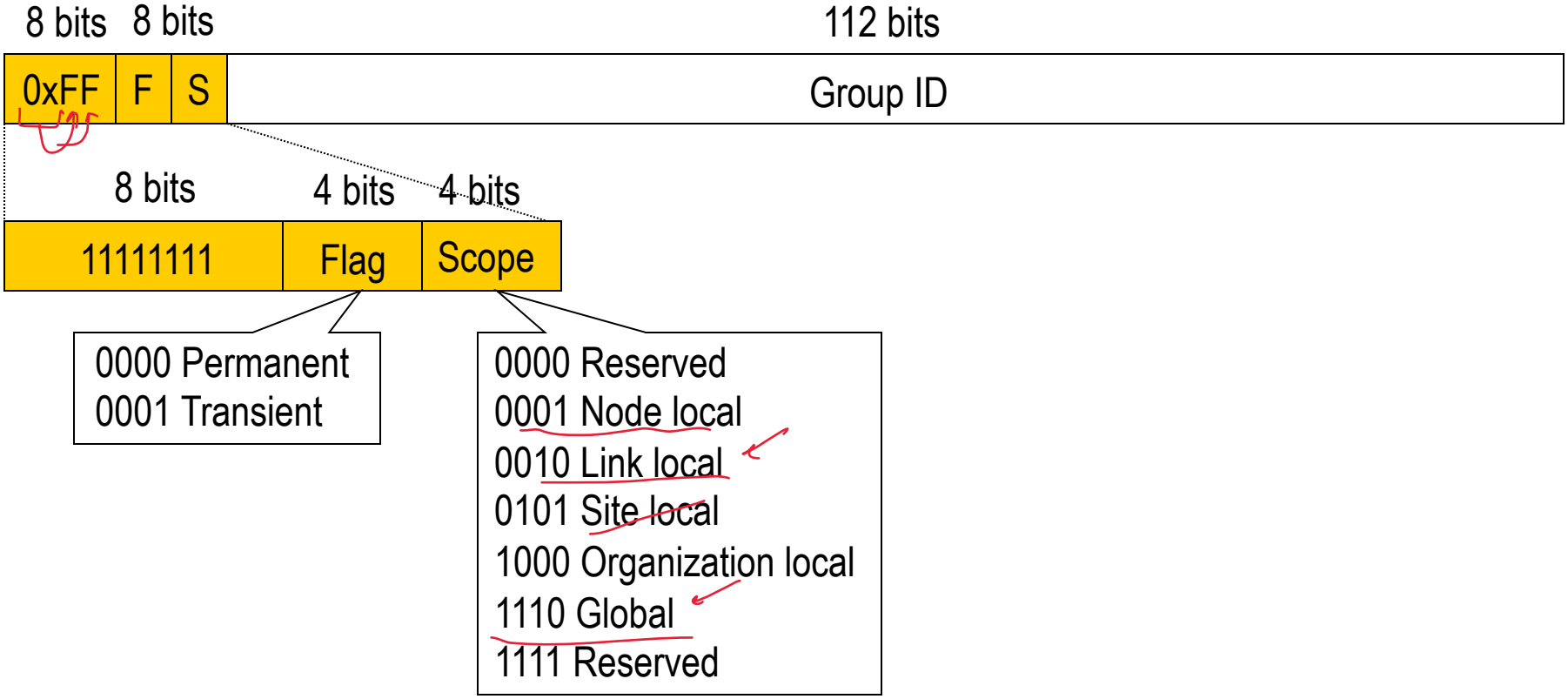
- The general format for IPv6 global unicast addresses is as follows




- where
  - **global routing prefix is a (typically hierarchically-structured) value assigned to a site (a cluster of subnets/links)**
  - **subnet ID is an identifier of a link within the site**
  - **interface ID uniquely identify interfaces on a link; it can be derived directly from that interface's link-layer address**
- All global unicast addresses other than those that start with binary 000 have a 64-bit interface ID field (e.g. n=48, m=16)
  - **the 64 bits interface ID is constructed in Modified EUI-64 format**
- Global unicast addresses that start with binary 000 have no such constraint on the size or structure of the interface ID field

# Multicast Addresses

**Multicast addresses**



# ICMPv6

- Internet Control Message Protocol per IPv6, used for:
  - **error reporting**
  - **diagnostic**
  - **neighbor discovery**
  - **management of multicast groups**
- It replaces some functions previously done (for IPv4) by:
  - **ICMP**
  - **ARP (Address Resolution Protocol)**
  - **IGMP (Internet Group Membership Protocol)**
- Two classes of messages:
  - **Error messages**
  - **Informational messages**
    - diagnostic
    - neighbor discovery
    - management of multicast groups

Sent by a router to indicate to a source node that it needs to fragment the packet

# ICMPv6 messages

- Error messages (types from 0 to 127)
  - **Destination Unreachable**
  - **Packet Too Big**
  - **Time Exceeded**
  - **Parameter Problem**
- Informational messages (types from 128 to 255)
  - **Echo Request**
  - **Echo Reply**
  - **Router Solicitation**
  - **Router Advertisement**
  - **Neighbor Solicitation**
  - **Neighbor Advertisement**
  - **Redirect**
  - **Group management (Multicast Listener Discovery Protocol)**

# Neighbor Discovery protocol

skip

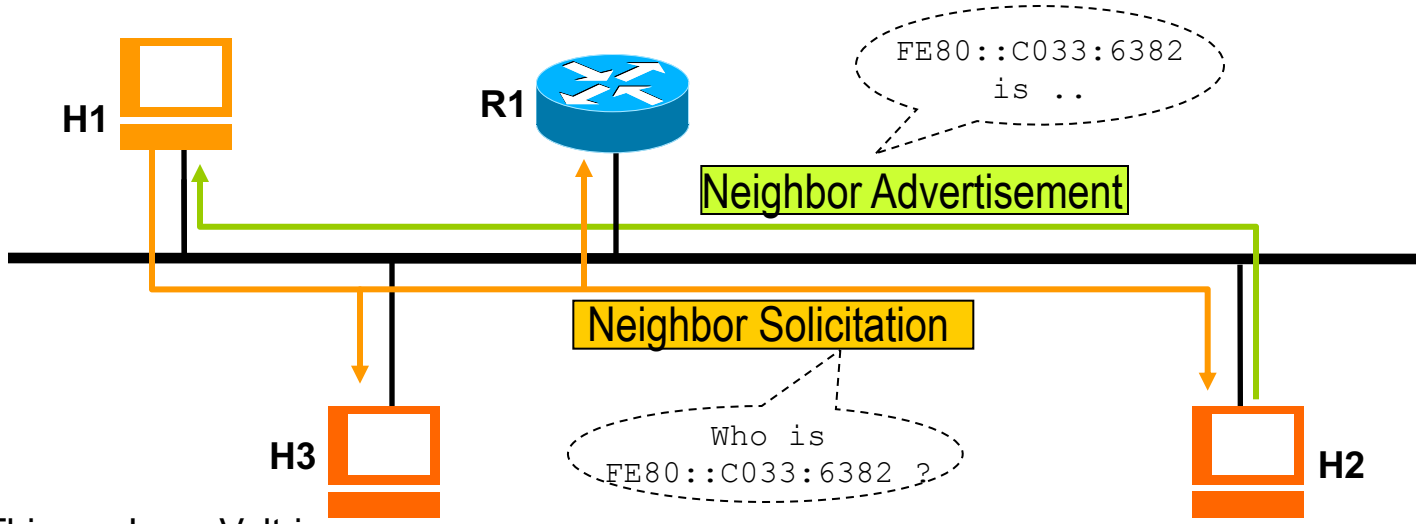
- It solves a set of problems related to the interaction between nodes attached to the same link
  - **Router Discovery**
    - how hosts locate routers that reside on an attached link
  - **Prefix Discovery**
    - how hosts discover the set of address prefixes that define which destinations are on-link for an attached link
  - **Parameter Discovery**
    - How a node learns such link parameters as the link MTU or such Internet parameters as the hop limit value to place in outgoing packets
  - **Address Autoconfiguration**
    - Introduces the mechanisms needed in order to allow nodes to automatically configure an address for an interface

# Neighbor Discovery protocol (cont.) Skip

- **Address resolution**
  - How nodes determine the link-layer address of an on-link destination (e.g., a neighbor) given only the destination's IP address
- **Next-hop determination**
  - The algorithm for mapping an IP destination address into the IP address of the neighbor to which traffic for the destination should be sent
  - The next-hop can be a router or the destination itself
- **Neighbor Unreachability Detection**
  - How nodes determine that a neighbor is no longer reachable
  - For neighbors used as routers, alternate default routers can be tried
- **Duplicate Address Detection**
  - How a node determines that an address it wishes to use is not already in use by another node
- **Redirect**
  - How a router informs a host of a better first-hop node to reach a particular destination

# Address Resolution

- The process through which a node determines the link-layer address of a neighbor given only its IP address
- Neighbor Solicitation
  - **Sent by a node when it has a unicast packet to send to a neighbor, but does not know the neighbor's link-layer address**
  - **Used also for Duplicate Address Detection**
- Neighbor Advertisement
  - **In reply to Neighbor Solicitation**
  - **Sent also for notifying a change of address**



# Host Autoconfiguration

- IPv6 supports different type of host auto-configurations
- Two “plug-and-play” ways for host configuration:
  - **Stateful**
    - DHCPv6 (Dynamic Host Configuration Protocol)
    - Similar to DHCP for IPv4
  - **Stateless**
    - Message exchange between Host and Router
    - Network prefix from ICMP Router Solicitation/Advertisement
    - The host forms its own address by appending its PH/DL interface address (e.g. MAC address)
- The two modes may coexist

# Multicast Listener Discovery Protocol (MLDv2)

- Multicast Listener Discovery Protocol (MLD)
- Used by IPv6 routers to
  - **discover the presence of multicast listeners (i.e., nodes that wish to receive multicast packets) on their directly attached links, and**
  - **discover specifically which multicast addresses are of interest to those neighboring nodes**
- MLDv2 is a translation of the IGMPv3 protocol for IPv6 semantics

# Transition mechanisms

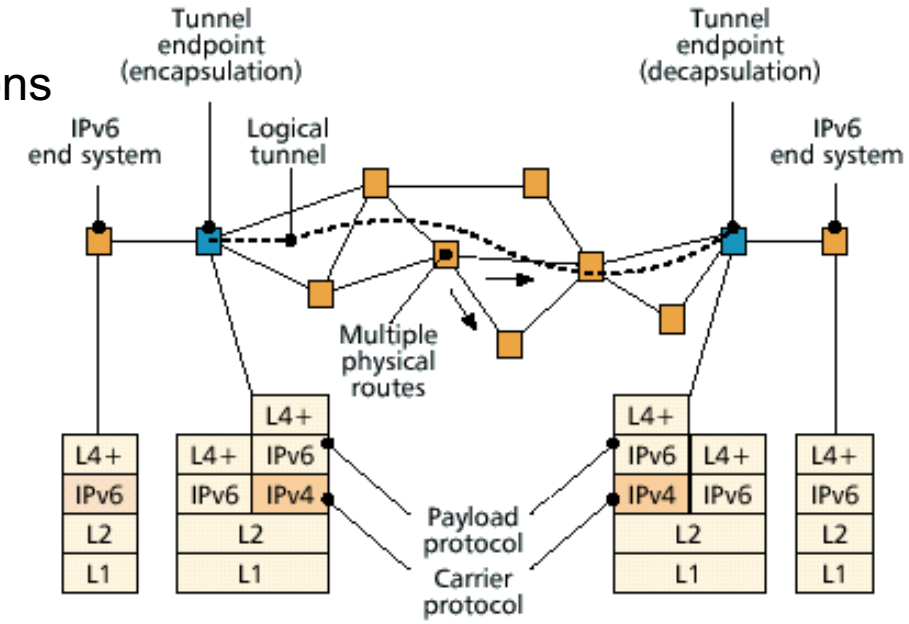
# Transition mechanisms

- Transition mechanisms generally come in one one three forms
  - **Dual stack**
  - **Translation**
  - **Tunneling**

- Dual stack
  - **two protocol stacks are separately maintained and operate in parallel**
  - **the device can operate via either protocols**
  - **can be implemented in both end systems and network nodes**
    - in end systems: it enables both IPv4 and IPv6 applications to operate on the same node
    - in network nodes: it allows handling of both IPv4 and IPv6 packets

# Transition mechanisms

- Tunneling
  - **used to bridge compatible nodes (e.g. IPv6) across incompatible networks (e.g. IPv4)**
  - **IPv6-over-IPv4 or IPv4-over-IPv6**
  - **main problem is the configuration of tunnels (on tunnel endpoints)**
  - **tunnel endpoint addresses are generally attained:**
    - manually or by tools (e.g. tunnel broker)
    - through existing services such as DNS or DHCP options
    - by embedding information in the link layer addresses or IP addresses



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# Communication Mechanisms

**Main sources**

- J. P. Vasseur and A. Dunkels, "Interconnecting Smart Objects with IP,` Morgan Kaufmann, 2010. Chapter 12.

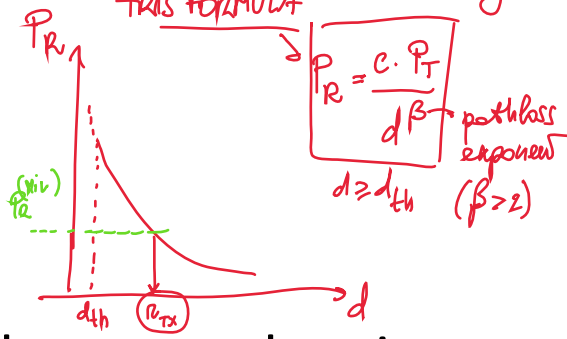
# Communication Patterns

- Smart object communication: one-to-one, one-to-many, and many-to-one.
- Each communication pattern is used in different situations. Many applications use a combination of the patterns.
- Smart objects often communicate over unreliable communication channels: (i) interference (shared medium and free spectrum band); (ii) obstacles on the path
- Need for reliable communication protocols (best-effort may be sufficient in some cases).
- Different requirements: static (e.g., industrial WSN) vs dynamic (BSN).
- Power consumption: switch off as much as possible.

# One-2-one Communication




**FIGURE 12.1**
One-to-one communication in a smart object network.




- One source and one destination, with possibly more relays in the middle (ad-hoc network)
- Could rely on the extensive body of knowledge on ad-hoc wireless networks

# One-2-many Communication

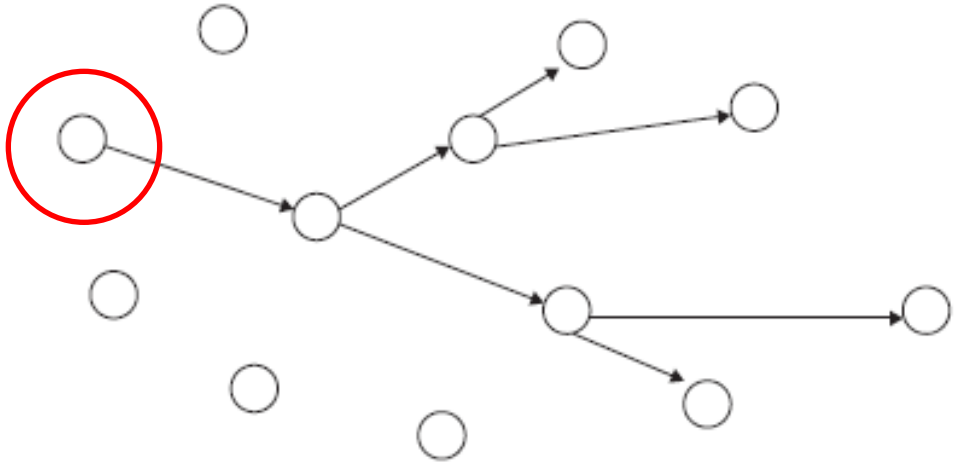


**FIGURE 12.2**
One-to-many communication in a smart object network.

- Need to reach all nodes (e.g., sending a command)
- Various mechanisms: flooding; irresponsible forwarding (IF); trickle (explicitly designed for low-power networks→RPL)
- Used in routing protocols for one-2-one communications (e.g., Ad-hoc On-demand Distance Vector, AODV) → iAODV

# Many-2-one Communication

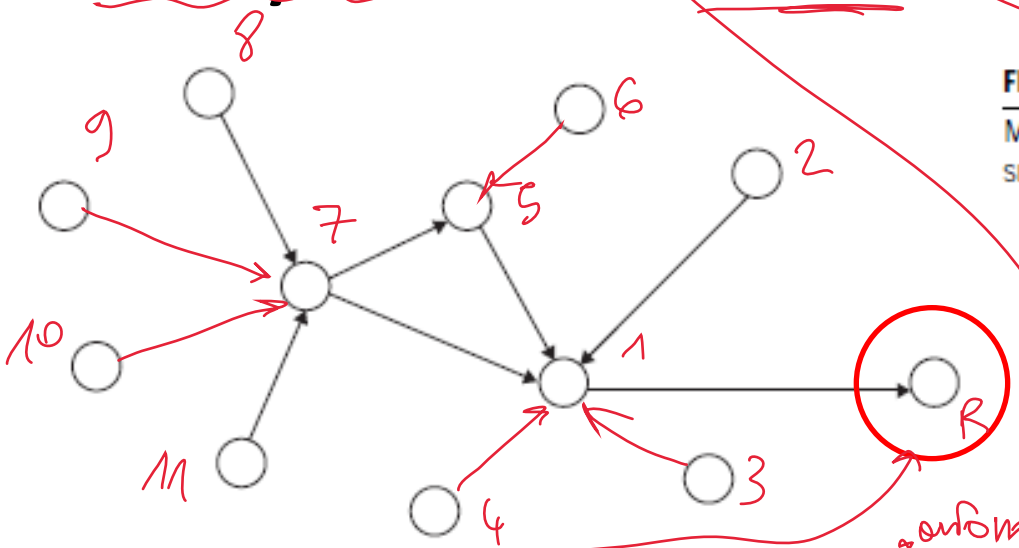
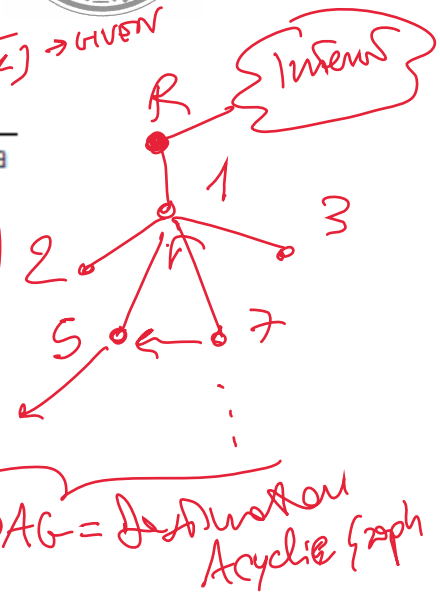



**FIGURE 12.3**
Many-to-one communication in a smart object network.




- Collect at a *sink*: sensor data or network health status (e.g., periodic status reports)
- There may be more than one sink inside the network.
- Set-up: build a tree structure with its root at the sink (iterative process from sink to all nodes) → hop count-based routing
- To account for radio quality in addition to hop count, several cost metrics can be considered → expected transmissions (ETX) is a good metric
- Collection Tree Protocol (CTP) based on ETX

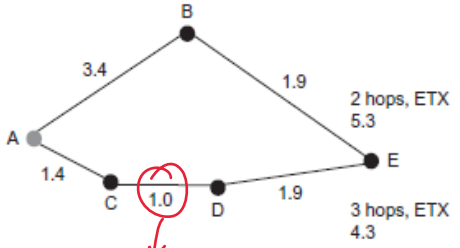



**FIGURE 12.4**
ETX in a five-node network.

# Physical Communication Standards

- Of interest: IEEE 802.15.4 (wireless), IEEE 802.11 (wireless), Power Line Communications (PLC, wired)
- Transmission ranges:
  - IEEE 802.15.4: short range (a few m)
  - IEEE 802.11: medium range (up to several hundred m)
  - PLC: potentially long range (depends on eletric loads connected to the cable)
- Network formation
  - PLC: all nodes connected to the same cable (same connectivity level)
  - IEEE 802.15.4: topology change (mobile nodes or environment change)
  - IEEE 802.11: similar to IEEE 802.15.4 (problems less pronounced)
- Routing
  - IEEE 802.15.4: routing is crucial (nodes act often as relays)
  - PLC: no real need for routing (broacast in the cable)
  - IEEE 802.11: for extended network coverage, APs are connected

# IEEE 802.15.4: Basics

- Developed by the 802.15 personal area network (PAN) Working Group within the Institute of Electrical and Electronics Engineers (IEEE).
  - Maximum data rate: 250 kb/s
  - Maximum transmit power: 1 mW
  - Nominal transmit range: a few tens of meters
- Focus: to allow low-cost and low-complexity transceivers
- Low-power radio stacks are built on IEEE 802.15.4: WirelessHART, ISA100a, IPv6, and ZigBee.
- The IEEE 802.15.4 standard specifies two layers: PHY and MAC (several mechanisms are proposed, but not all are used)
- Maximum PHY packet (payload) size: 127 bytes → data for upper layers is between 86 and 116 bytes (besides the MAC header) → upper layers need to fragment large data portions into 802.15.4 frames
- IEEE 802.15.4 is typically implemented as a combination of hw (PHY and part of MAC) and sw (MAC logic)

# IEEE 802.15.4: Topologies

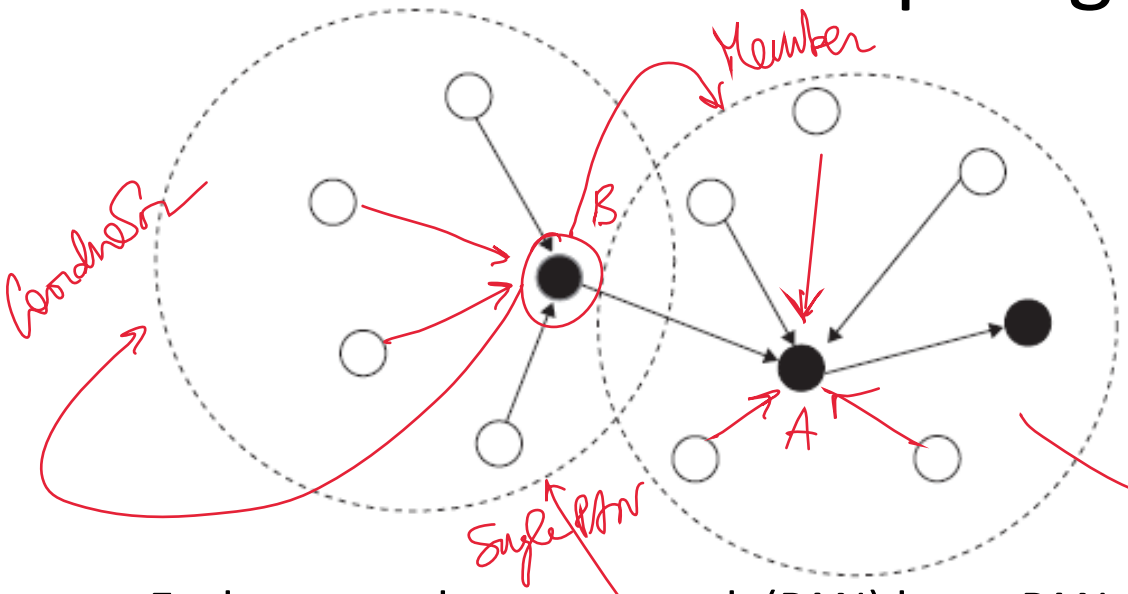
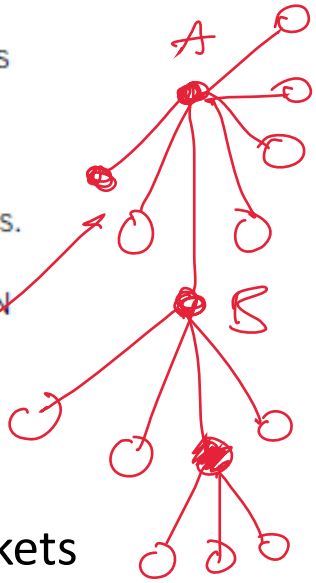


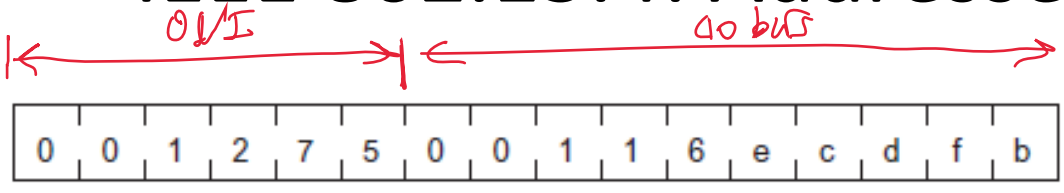

**FIGURE 12.5**

An IEEE 802.15.4 network with FFDs shown as dark dots and RFDs shown as hollow dots. Two of the FFDs are PAN coordinators in the two PANs, shown as dotted circles. The right PAN contains two FFDs, but only one is the PAN coordinator.




- Each personal area network (PAN) has a PAN coordinator and packets carry a 16-bit identifier.
- A device can participate in one PAN as the PAN coordinator and simultaneously participate as a PAN member in another PAN.
- Two types of nodes: fully functional devices (FFDs, can be coordinators) and reduced function devices (RFDs, cannot be coordinators)
- Three types of network topology: star, mesh, and cluster tree
- Routing is minor → protocols on top of 802.15.4 build their own topology and routing protocols

# IEEE 802.15.4: Addresses



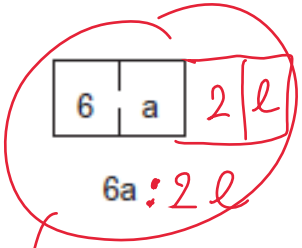




**FIGURE 12.6**
IEEE 802.15.4 supports two addressing formats: long (64-bit) addresses and short (16-bit) addresses.

MAC address ! fixed

- Each node in an 802.15.4 network has a 64-bit address that uniquely identifies the device: very long!
- Long addresses are globally unique and each 802.15.4 device is assigned an address when manufactured (24-bit for organizational unique identifier (OUI) from the IEEE; 40 bit assigned by the manufacturer), To the specific producer
- Short addresses (16-bit) are assigned (run-time) by the PAN coordinator and are valid only within the context of a PAN

# IEEE 802.15.4: PHY layer (1/2)

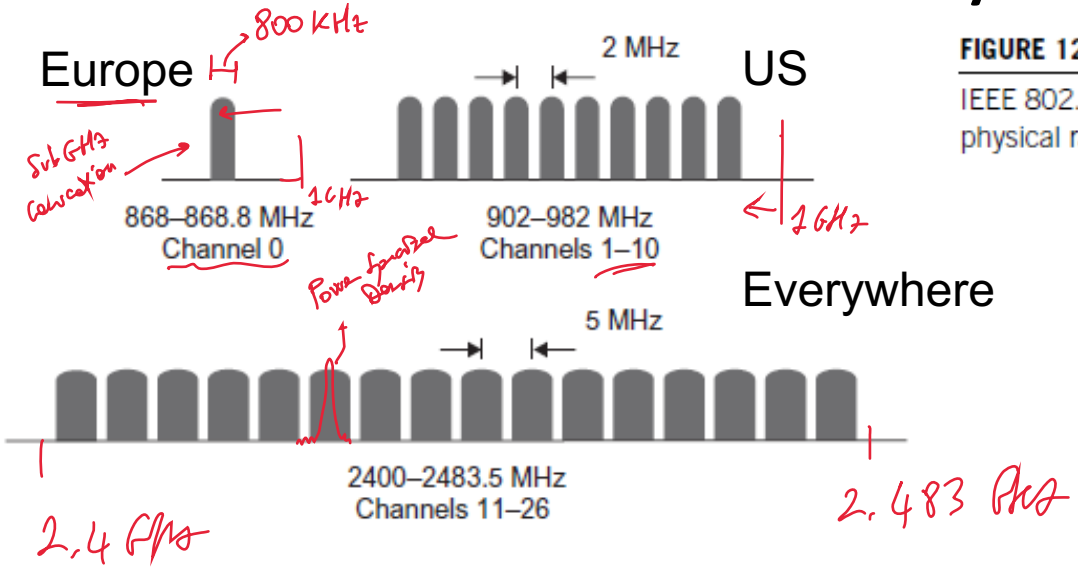



**FIGURE 12.7**

IEEE 802.15.4 defines 26 physical radio channels.

- **Modulations:**
  - Channels 0-10: BPSK
  - Channels 11-26: QPSK
- **Bit rates:**
  - ch. 0: 20 kb/s
  - ch. 1-10: 40 kb/s
  - ch. 11-26: 250 kb/s

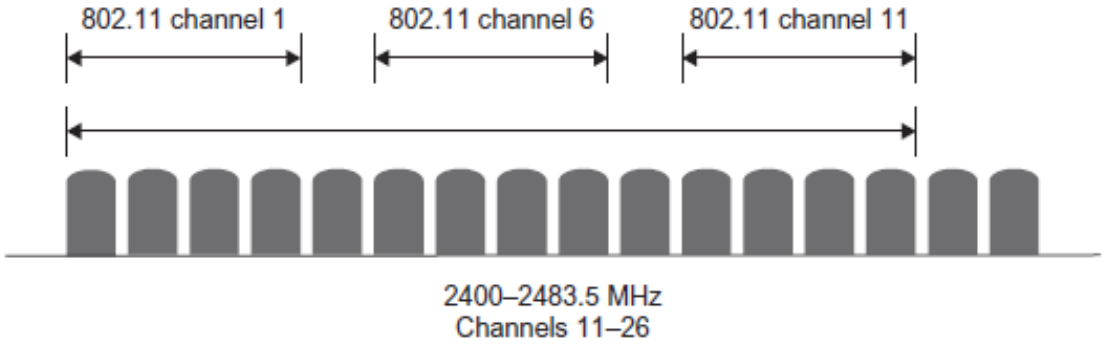
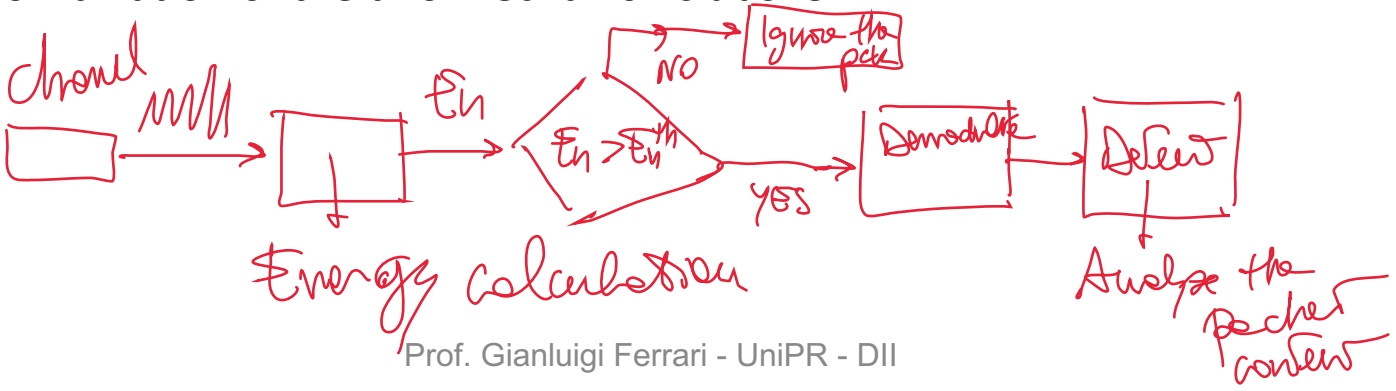



**FIGURE 12.8**

IEEE 802.15.4 channels 11–24 overlap the 802.11 channels. Channels 25 and 26 are not covered by 802.11 channels. When the non-overlapping 802.11 channels 1, 6, and 11 are used, two additional 802.15.4 channels are undisturbed by 802.11.

# IEEE 802.15.4: PHY layer (2/2)

- Interference from IEEE 802.11 devices (WiFi) is a serious issue → lossy links
- PHY layer: provides mechanisms to measure the radio energy for a given radio channel: clear channel assessment (CCA) → used by MAC layer
- CCA uses three mechanisms:
  - measure the radio energy and compare it with a predefined energy threshold
  - demodulate the incoming radio signal to see if it is a valid 802.15.4 signal
  - a combination of the two mechanisms above



# IEEE 802.15.4: MAC layer

- MAC layer: control access to the radio medium → control when the medium is idle (CCA) and how to coordinate access by multiple nodes
- IEEE 802.15.4 MAC layer provides: channel access management, validation of incoming frames, and acknowledgment of frame reception.
- Optional mechanisms for a time-division multiple access (TDMA) mechanism for medium access: the PAN coordinator assigns time slots to PAN devices (beacon mode: typically not of interest)
- Validation of incoming frames by computing: 16-bit cyclic redundancy check (CRC) of the entire frame
- If an incoming frame has the acknowledgment bit set → MAC layer broadcasts an acknowledgment frame → interference!

# IEEE 802.15.4: Frame formats

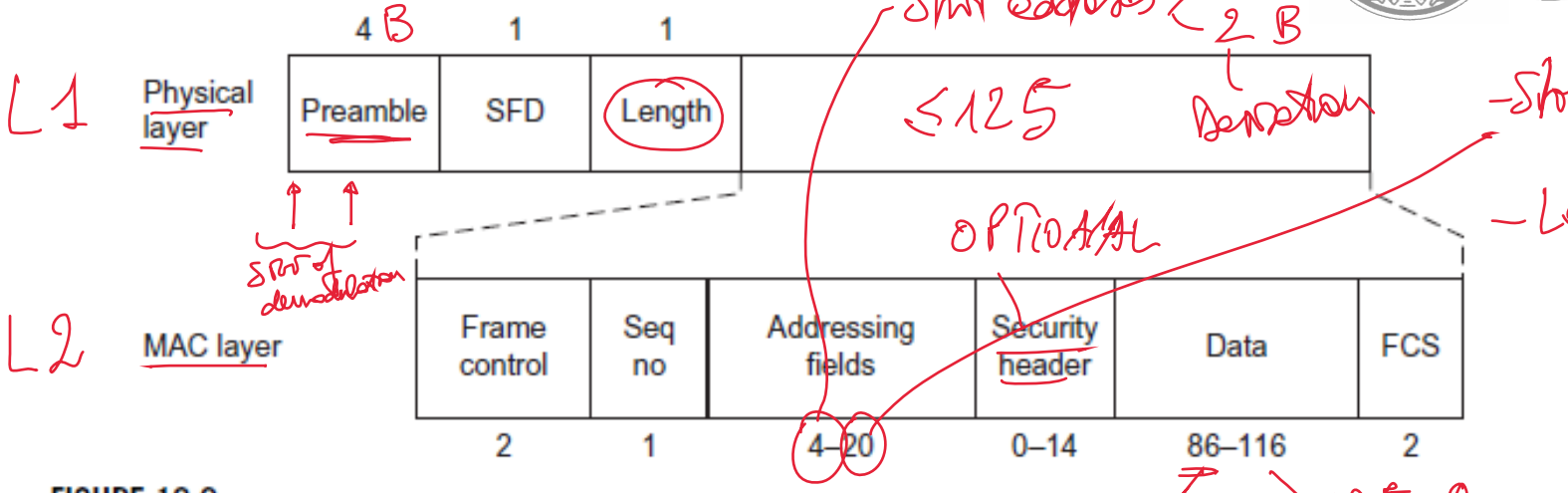



**FIGURE 12.9**

The IEEE 802.15.4 physical layer and MAC layer header formats.

- **PHY frame**:
  - Preamble: used to synchronize
  - Start of Frame Delimiter (SFD): indicates end of preamble
  - Length: between 0 and 2^7-1=127
- **MAC frame**:
  - Frame control: flags to interpret the rest of the header and request of ack
  - Sequence number: used by ack
  - Addressing fields: optional (sender/receiver, PAN)
  - Frame check sequence (FCS): contains the CRC to be checked

# IEEE 802.15.4: Power consumption

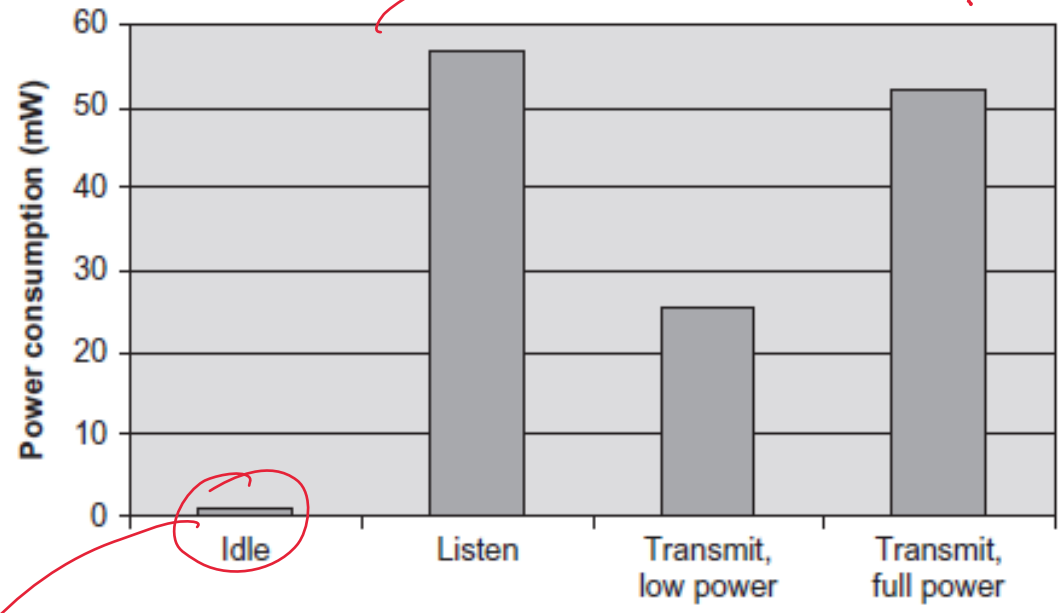



**FIGURE 12.10**

The power consumption of the CC2420 IEEE 802.15.4 radio transceiver.

- Listen power consumption higher than transmit power consumption (needs to demodulate even if no useful signal is present)
- Transmit power can be set on a per-packet basis (between low and full)

# The ISO/OSI Stack of IEEE 802.15.4

